SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 2 – Agile Sprint Planning & User Story Workshop

Course Code:	CSA10 / CSA1001	**Course Title:**	Software Engineering
CO Assessed:	CO1	**Assessment:**	Assessment Tool 2 – **Agile Sprint Planning & User Story Workshop**
Weightage:	20%	**Total Marks:**	25
CO1 Statement:	*Apply advanced methodologies and agile project management techniques to manage software projects effectively*		
Student Name:	S Surender	**Reg. No.:**	1924260070
Date:		**Duration:**	60 minutes

Code of Conduct

I,S SURENDER(1924260070) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: S SURNEDER

Annexure A – Agile Sprint Planning & User Story Workshop

Instructions to Students:

Each student is assigned an individual software project problem statement. Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

Smart Hospital Appointment and Patient Queue Management System (SHAQMS).

1. Identify the Project Vision and Stakeholders

Define the project's vision, objectives, and identify all key stakeholders involved.

1.1 Project Vision

To develop a secure and user-friendly hospital appointment and patient queue management platform that simplifies doctor availability checks, appointment booking, and real-time queue updates. SHAQMS aims to reduce waiting uncertainty, prevent double bookings, support reception staff, and improve coordination between patients, doctors, and hospital administrators.

1.2 Project Objectives

The main objectives of the Smart Hospital Appointment and Patient Queue Management System (SHAQMS) are:

- To enable patients to view available doctors, departments, and appointment slots online.
- To allow patients to book, reschedule, and cancel appointments conveniently.
- To prevent double bookings through real-time slot availability validation.

- To generate unique queue tokens for confirmed appointments and walk-in registrations.
- To display current queue status and estimated waiting information on a centralized dashboard.
- To notify patients about appointment confirmation, changes, and queue progress.
- To help receptionists register arrivals and update check-in status efficiently.
- To provide doctors with an organized view of upcoming appointments and waiting patients.
- To store appointment and queue records securely with role-based access.
- To reduce administrative workload and improve patient experience through accurate scheduling and transparent queues.

1.3 Key Stakeholders and Responsibilities

Stakeholder	Role / responsibility
Hospital administrator	Configures doctors, schedules, accounts, and operational settings.
Patients	Searches doctor availability, books visits, and views queue updates.
Doctors	Reviews appointments and coordinates consultation flow.
Reception staff	Checks in patients, assigns tokens, and advances queues.
Nursing staff	Coordinates patient movement and supports queue operations.
Hospital management	Reviews appointment load, wait times, and service quality.
Development and QA team	Builds, tests, secures, and maintains the platform.

1.4 Scope and Assumptions

The first release covers doctor schedule registration, appointment booking, queue token generation, check-in, and basic queue status updates. Electronic medical records, billing, and emergency triage are outside the first sprint. Appointment records are assumed to include patient identifier, doctor, date, time slot, status, and token.

2. Create Epics and User Stories

Break down the requirements into Epics and formulate well-structured User Stories using the format: As a <user>, I want <feature>, so that <benefit>.

2.1 Epic: Appointment Booking and Availability

US-01: As a patient, I want to view available doctor appointment slots, so that I can select a suitable consultation time.

US-02: As a patient, I want to book or cancel an appointment, so that I can manage my visit without unnecessary travel.

2.2 Epic: Queue and Check-in Management

US-03: As a receptionist, I want to check in arriving patients and assign queue tokens, so that visits are organized in arrival order.

US-04: As a patient, I want to see my queue position and current token status, so that I can plan my waiting time.

2.3 Epic: Notifications and Reporting

US-05: As a patient, I want to receive appointment and queue notifications, so that I do not miss important updates.

US-06: As a hospital manager, I want to review daily appointment and waiting-time summaries, so that I can identify scheduling bottlenecks.

2.4 Epic: Hospital Administration

US-07: As an administrator, I want to register doctors and configure their consultation slots, so that patients can book only valid appointments.

US-08: As an administrator, I want to assign staff permissions, so that patient and scheduling information is accessed only by authorized users.

3. Define Acceptance Criteria

Write clear and measurable acceptance criteria for each selected user story using the Given–When–Then format or equivalent.

3.1 US-01 – Doctor Slot Availability

Given a doctor has published consultation slots, when a patient selects the doctor and date, then the system displays only available slots. Given a slot is already booked, when another patient checks availability, then that slot is not offered for booking.

3.2 US-03 – Patient Check-in and Token

Given a patient has a confirmed appointment, when reception checks the patient in, then the system assigns a unique queue token and records the check-in time. Given the patient is already checked in, when the action is repeated, then no duplicate token is created.

3.3 US-04 – Live Queue Status

Given a checked-in patient has an active token, when reception advances the queue, then the patient dashboard shows the updated current token and queue position within 10 seconds. Given the appointment is cancelled, when the queue refreshes, then the cancelled token is not shown as waiting.

3.4 US-07 – Doctor Schedule Registration

Given an authorized administrator is signed in, when a doctor and non-overlapping consultation slots are submitted, then the schedule is saved and displayed. Given a new slot overlaps an existing slot for the same doctor, when it is submitted, then the system rejects it with a clear message.

4. Prioritize the Product Backlog

Organize and prioritize the product backlog based on business value, customer needs, and project dependencies.

4.1 Prioritized Product Backlog

Rank	Story	Priority	Reason/ dependency
1	US-07 Doctor schedule	Must	Required to publish valid consultation slots.
2	US-01 Slot availability	Must	Patients need reliable schedule visibility.
3	US-03 Patient check-in	Must	Required to generate active queue tokens.
4	US-04 Live queue status	Must	Depends on checked-in tokens.
5	US-08 Staff permissions	Should	Protects access to patient and staff functions.
6	US-02 Appointment booking	Should	Builds on validated slot availability.
7	US-05 Notifications	Could	Useful after appointment and queue events exist.
8	US-06 Daily reports	Could	Depends on stored appointment history.

4.2 Prioritization Method

MoSCoW prioritization is used: Must-have features establish doctor availability, booking, check-in, and queue visibility; Should-have features improve notifications and access control; Could-have features add management reporting. Dependencies are considered before sprint selection.

5. Plan the Sprint Backlog

Select high-priority user stories for the first sprint and divide them into actionable development tasks.

5.1 Sprint 1 Overview

Sprint duration: 2 weeks (10 working days). Team assumption: four student contributors. Sprint 1 includes US-07, US-01, US-03 and US-04 using sample doctor and patient records for demonstration.

5.2 Sprint Backlog and Development Tasks

Story	Actionable tasks	Expected result
US-07	Create doctor and slot data model; build schedule form; reject overlapping slots; test saving and retrieval.	Administrator can publish valid doctor schedules.
US-01	Build availability API; filter booked slots; design patient slot list; test date and doctor filters.	Patients can view available slots.
US-03	Build patient check-in screen; generate unique tokens; prevent duplicate check-ins; store timestamps.	Reception can register arrivals and issue tokens.
US-04	Create queue API; display current token and waiting position; update on queue advance; test refresh.	Patients can see updated queue progress.

5.3 Sprint Workflow and Definition of Done

Sprint Workflow:

The Smart Hospital Appointment and Patient Queue Management System follows an Agile sprint workflow to deliver reliable scheduling and transparent patient queues through iterative development and testing.

- Sprint Planning: Identify high-priority user stories and assign development tasks to team members.
- Requirement Analysis: Identify doctor availability, appointment booking, patient check-in, and queue tracking requirements.
- System Design: Design patient and receptionist interfaces, appointment database tables, and queue token rules.
- Development: Implement doctor schedules, available slot display, check-in, and queue status updates.
- Testing: Verify schedule conflicts, token uniqueness, check-in actions, and live queue updates.
- Sprint Review: Demonstrate completed features and collect feedback from stakeholders.

- Sprint Retrospective: Evaluate team performance, identify challenges, and suggest improvements for the next sprint.

Definition of Done (DoD):

A user story is considered completed when the following conditions are satisfied:

- All specified functional requirements are implemented successfully.
- The developed features satisfy the defined acceptance criteria.
- Doctor availability and appointment slots are displayed correctly.
- Patient check-in generates unique queue tokens without duplicates.
- The dashboard displays current queue position and token status.
- Appointment and queue data is stored securely in the database.
- The system is tested to identify and resolve errors.
- The completed functionality is reviewed and approved by the project team.
- All necessary technical documentation is updated.
- The completed feature is ready for demonstration and integration with other system modules.

6. Estimate Effort Using Story Points

Assign story points to each selected user story using an appropriate estimation technique (e.g., Fibonacci sequence or Planning Poker) and justify your estimates.

6.1 Estimation Technique

Story points are used in Agile software development to estimate the relative effort required to complete user stories. The estimation considers task complexity, development effort, technical challenges, and potential risks.

In SHAQMS, story points help the development team estimate scheduling complexity, queue coordination, testing effort, and sprint capacity.

6.2 Story Point Estimates

Story	Points	Justification
US-07	3	Schedule form, validation, storage, and overlap checks.
US-01	5	Availability filtering, doctor/date selection, and testing.
US-03	5	Check-in workflow, unique token generation, and edge cases.
US-04	3	Queue display, refresh, and status transitions.

Total	16	Proposed Sprint 1 effort; confirm through team estimation.

6.3 Estimation Risks

User Story	Development Task	Story Points
US01	User login and authentication	3
US02	Doctor availability and schedules	5
US03	Appointment booking and cancellation	5
US04	Queue token management	8
US05	Appointment and queue notifications	5
US06	Live queue dashboard	5
US07	Appointment and queue data storage	3
US08	Hospital appointment reports	5
Total		39

6.4 Justification of Story Point Estimation

- User Authentication (3 Points): Requires login validation, user roles, and basic access security.
- Doctor Availability (5 Points): Requires doctor schedules, slot filtering, and availability display.
- Appointment Booking (5 Points): Requires slot reservation, cancellation, and conflict prevention.
- Queue Token Management (8 Points): Requires token assignment, ordering rules, and duplicate prevention.
- Appointment Notifications (5 Points): Requires event triggers and reliable patient status messages.
- Queue Monitoring Dashboard (5 Points): Requires live token and queue position updates.
- Appointment Data Storage (3 Points): Requires database connectivity and reliable record retrieval.
- Hospital Reports (5 Points): Requires daily summaries and appointment trend calculations.

7. Present the Sprint Goal and Expected Deliverables

Define a clear sprint goal and describe the expected sprint deliverables, including the features to be completed at the end of the sprint.

7.1 Sprint Goal

The main goal of the first sprint is to implement doctor schedule registration, available slot display, patient check-in, and a basic live queue dashboard. These features establish the foundation for reliable hospital appointment and patient queue management.

The sprint aims to reduce scheduling confusion, prevent duplicate tokens, and provide clear visibility into patient waiting order.

7.2 Expected Deliverables

Deliverables include: a doctor schedule administration screen; available appointment slot display; patient check-in form; unique queue token generation; live queue status dashboard; appointment and token database records; acceptance test results; and a short demonstration using sample data.

7.3 Sprint Review and Success Measures

The sprint is successful when all four selected stories satisfy their acceptance criteria, overlapping doctor slots are rejected, queue tokens are unique, and the displayed queue updates correctly. Stakeholder feedback will guide backlog refinement.

7.4 Proposed Next Sprint

Sprint 2 can add full appointment booking and cancellation, patient notifications, role-based permissions, and daily waiting-time reports, depending on team velocity and feedback.

Conclusion

The Agile plan translates SHAQMS into measurable, prioritized development work. By focusing the first sprint on doctor schedule registration, slot visibility, check-in, and queue tracking, the project establishes a practical foundation for reducing waiting uncertainty and improving hospital coordination. Future sprints can expand patient self-service and operational reporting.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Project Vision & Stakeholder Identification (30%)	Project vision is clear, comprehensive, and aligned with the problem statement. All relevant stakeholders and their roles are correctly identified.	Vision is clear with minor omissions. Most stakeholders are correctly identified.	Vision is partially defined. Some stakeholders are missing or incorrectly identified.	Vision is unclear or incomplete. Stakeholders are poorly identified or missing.
Epics & User Stories (25%)	Epics are well-defined and user stories are complete, follow Agile format, and fully capture functional requirements.	Epics and user stories are mostly complete with minor formatting or requirement issues.	User stories are partially complete with limited Agile structure.	User stories are incomplete, unclear, or do not follow Agile practices.
Acceptance Criteria (20%)	Acceptance criteria are specific, measurable, testable, and aligned with all user stories.	Acceptance criteria are mostly clear and testable with minor gaps.	Acceptance criteria are partially defined and lack clarity or completeness.	Acceptance criteria are missing or not aligned with user stories.
Product Backlog Prioritization & Sprint Planning (15%)	Product backlog is logically prioritized, sprint backlog is realistic, and task breakdown is complete.	Backlog prioritization and sprint planning are mostly appropriate with minor issues.	Prioritization or sprint planning is partially complete with limited justification.	Backlog prioritization and sprint planning are poorly organized or incomplete.
Story Point Estimation, Sprint Goal & Presentation (10%)	Story point estimation is well justified, sprint goal is clear, deliverables are realistic, and presentation is professional and well organized.	Estimation and sprint goal are mostly appropriate with minor inconsistencies. Presentation is clear.	Estimation or sprint goal lacks justification. Presentation is adequate but incomplete.	Estimation is unrealistic or missing. Sprint goal, deliverables, and presentation are unclear or incomplete.

Criteria	Mark
Project Vision & Stakeholder Identification **(30%)**	/5
Epics & User Stories **(25%)**	/5
Acceptance Criteria **(20%)**	/5
Product Backlog Prioritization & Sprint Planning **(15%)**	/5
Story Point Estimation, Sprint Goal & Presentation **(10%)**	/5
TOTAL	/25